

## WEB TECHNOLOGIES LAB

### III B. TECH- I SEMESTER

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A5CS12      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | -            | - | 3 | 1.5     | 30            | 70  | 100   |

#### COURSE OBJECTIVES

1. Teach Students to build static web pages using HTML and CSS
2. Explain web application development procedures
3. Facilitate students to build dynamic, responsive web pages using Angular JS
4. Teach students the basics of server side scripting using PHP
5. Familiarize various concepts Java Framework Struts.

#### COURSE OUTCOMES

1. Use HTML and CSS to design web pages.
2. Infer the role of XML in web applications and use them.
3. Apply the concepts of JavaScript to perform client side validation and create dynamic web pages.
4. Create server side applications using PHP.
5. Create web application using AngularJS that depend on MVC architecture.
6. Applying the concept of Struts framework.

#### SYLLABUS

##### Note:

1. Apache, MySQL and PHP for the Lab Experiments. Though not mandatory, encourage the use of Eclipse platform wherever applicable
  2. The list suggests the minimum program set. Hence, the concerned staff is requested to add more problems to the list as needed
1. Install the following on the local machine
    - Apache Web Server (if not installed)
    - Tomcat Application Server locally
    - Install MySQL (if not installed)
    - Install PHP and configure it to work with Apache web server and MySQL (if not already configured)
  2. Write a HTML Page to create a Student registration form that collects various student information such as first name, last name, Roll No., email id, mobile number, etc and a two buttons: submit and reset. Apply different CSS properties
  3. Implement the following web application using XML. A user validation web application, where the user submits the login name and password to the server. The name and password are checked against the data already available in the XML file and if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.
  4. Create an XMLHttpRequest to retrieve data from an XML file and display the data in an HTML table.
  5. Write an HTML page that contains a selection box with a list of 5 countries. When the user selects a country, its capital should be printed next to the list. Add CSS to customize the properties of the font of the capital (color, bold and font size).
  6. Write an HTML page that has one input, which can take multi-line text and a submit button. Once the user clicks the submit button, it should show the number of characters, words and lines in the text entered using an alert message. Words are separated with while space and lines are separated with new line character.
  7. Implement the following web applications using AngularJS: A simple calculator web application that takes two numbers and an operator (+, -, /, \* and %) from an HTML page and returns the

result page with the operation performed on the operands

8. Create a Angular JS Application to make a shopping list where you can add or remove items.
9. Implement the following web applications using AngularJS: A user validation web application, where the user submits the login name and password to the server. The name and password are checked if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.
10. Write an HTML page including any required PHP that takes a number from one text field in the range of 0 to 999 and shows it in range of 0 to 999 and shows it in another text field in words. If the number is out of range, it should show "out of range" and if it is not a number, it should show "not a number" message in the result box.
11. Implement the following web applications using PHP: A user validation web application, where the user submits the login name and password to the server. The name and password are checked against the data already available in Database and if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.
12. A web application that lists all cookies stored in the browser on clicking "List Cookies" button. Add cookies if necessary.

### TEXT BOOKS

1. Web Technologies, Uttam K Roy, Oxford University Press
2. The Complete Reference PHP – Steven Holzner, Tata McGraw-Hill

### REFERENCE BOOKS

1. Web Programming, building internet applications, Chris Bates 2<sup>nd</sup> edition, Wiley Dremtech
2. Java Server Pages – Hans Bergsten, SPD O'Reilly
3. Java Script, D.Flanagan, O'Reilly, SPD.
4. Beginning Web Programming-Jon Duckett WROX.
5. Programming world wide web, R.W. Sebesta. Fourth Edition, Pearson.
6. Internet and World Wide Web – How to program, Dietel and Nieto, Pearson

### WEB LINKS

1. <https://www.w3schools.com/php/>
2. <https://angularjs.org/>
3. [https://www.w3schools.com/angular/angular\\_intro.asp](https://www.w3schools.com/angular/angular_intro.asp)
4. <https://www.tutorialspoint.com/angularjs/index.htm>